

Graduate Programs — Aerospace Engineering

Programs are organized by research strength in the areas most relevant to a physics background. "Best for" reflects what each program does distinctively well, not a general ranking.

****Note:**** Verify faculty positions and active advising before applying. AE is a smaller field than physics and faculty moves matter.

Programs most aligned with a physics background

MIT — Department of Aeronautics and Astronautics (AeroAstro)

Best for: Electric propulsion, space systems, GNC, autonomy, astrodynamics

The broadest and deepest AE department in the country. Strong across nearly every sub-field. The Space Propulsion Lab works on electric propulsion — Hall thrusters, ion engines, plasma diagnostics. The Space Systems Lab builds and flies actual satellites. The AeroAstro program is large enough to find your direction in and small enough to know your colleagues.

Key faculty:

- Paulo Lozano — Electric propulsion, ion electrospray thrusters, miniaturized propulsion for CubeSats
- Richard Binzel — Planetary science / near-Earth objects (more science than engineering but exists here)
- Kerri Cahoy — Smallsats, laser communications, space systems

Notes: Most competitive AE program in the US alongside Caltech. Stipends are around \$40–44k but Boston is expensive. Being at MIT means proximity to the broader Boston research ecosystem — Harvard, Draper Lab, Lincoln Lab.

Princeton — Department of Mechanical and Aerospace Engineering (MAE)

Best for: Electric propulsion (world-class), plasma physics applications, astrodynamics

The Electric Propulsion and Plasma Dynamics Lab (EPPDyL) — founded by Edgar Choueiri — is one of the most distinguished plasma propulsion groups in the world. Choueiri retired recently but the group continues. Princeton also has strong astrodynamics work.

Key faculty:

- Christopher Smoot — Plasma propulsion, Hall thrusters
- Naomi Ehrich Leonard — Control theory, collective behavior, formation dynamics (autonomy)
- Jeremy Kasdin (may have retired/moved — verify) — Astrodynamics, space telescopes

Notes: Princeton MAE is smaller than MIT AeroAstro. The campus is in suburban New Jersey — comfortable but not a major city. Strong physics culture bleeds across from the physics department, which is a genuine advantage for someone coming from that background.

University of Michigan — Department of Aerospace Engineering

Best for: Electric propulsion, space weather, plasma physics applications, broad AE

Michigan's Plasmadynamics and Electric Propulsion Laboratory (PEPL) is consistently one of the top electric propulsion research groups in the country. The space weather group at Michigan is also strong — directly relevant to space environment research.

Key faculty:

- Benjamin Jorns — Hall thruster physics, plasma diagnostics, plume characterization
- Alec Thomas — Laser-plasma physics, high-power lasers (more physics-adjacent)
- Mark Kushner — Plasma engineering and modeling

Notes: Ann Arbor is a genuinely livable college town — affordable, good food scene, strong university culture. Michigan's AE department is one of the largest in the country, which means resources and variety. Stipends are moderate; Ann Arbor cost of living is low enough that they go far.

University of Colorado Boulder — Ann and H.J. Smead Dept of Aerospace Engineering Sciences

Best for: Astrodynamics (exceptional), space weather, space systems, cislunar dynamics

CU Boulder has an unusually concentrated strength in space — the Laboratory for Atmospheric and Space Physics (LASP) is one of the world's leading space science institutions, and the astrodynamics group is consistently top-ranked nationally. If astrodynamics pulls, Boulder deserves serious consideration.

Key faculty:

- Hanspeter Schaub — Spacecraft attitude control, formation flying, electrostatic charging
- Daniel Scheeres — Astrodynamics, small body dynamics, spacecraft-asteroid interactions
- Zach Putnam — Entry, descent, and landing; trajectory optimization

Notes: Boulder is one of the best places to live in the US — mountains, outdoor culture, strong scientific community (NCAR, NIST, NOAA all based here), genuinely beautiful. Stipends are moderate; cost of living is rising but still better than coastal cities. The department has a distinct space-first culture.

California Institute of Technology — GARCIT (Graduate Aerospace Laboratories)

Best for: Propulsion (chemical and advanced), high-speed flows, computational fluid dynamics, fundamental research

GARCIT is small, research-intensive, and extremely selective. It has a long tradition in propulsion — JPL was literally founded by Caltech researchers. The research is more fundamental than applied; GARCIT is not primarily a systems engineering department.

Key faculty:

- Guillaume Blanquart — Combustion, turbulence, soot formation
- Beverley McKeon — Turbulence, wall-bounded flows
- Morteza Gharib — Fluid mechanics, biological flows, sensing

Notes: Caltech is in Pasadena, adjacent to JPL. Very small graduate program — this is either a feature or a bug depending on your temperament. Extremely competitive. The physics-AE boundary is thin here.

Purdue University — School of Aeronautics and Astronautics

Best for: Propulsion (chemical and electric), astrodynamics, structures, broad AE

Purdue has one of the oldest and most comprehensive AE programs in the country. Strong propulsion tradition — many Apollo-era engineers trained here. The Zucrow Labs for high-pressure combustion are world-class. Also strong in astrodynamics and GNC.

Key faculty:

- Stephen Heister — Chemical propulsion, liquid rocket engines, combustion instability
- Kathleen Howell — Astrodynamics, three-body problem, libration point orbits (directly relevant to cislunar missions)
- Alina Alexeenko — Rarefied gas dynamics, microthrusters

Notes: West Lafayette, Indiana is a college town — affordable but limited beyond the university. Large program with significant resources. Purdue graduates are highly recruited by NASA, SpaceX, Blue Origin.

Georgia Institute of Technology — Daniel Guggenheim School of Aerospace Engineering

Best for: Hypersonics, propulsion, structures, broad AE, industry connections

Georgia Tech's AE program is one of the largest and best-resourced in the country. Particularly strong in hypersonics and high-speed aerodynamics. The Ben T. Zinn Combustion Lab is a leading propulsion facility.

Key faculty:

- Vigor Yang — Combustion, liquid rocket engines, energetic materials
- Graeme Kennedy — Structural optimization, composite structures
- Lakshmi Sankar — Rotorcraft aerodynamics, wind energy

Notes: Atlanta is a major city — real urban infrastructure, affordable cost of living, good food scene. Georgia Tech has very strong industry relationships (Lockheed Martin, Boeing, Delta, Raytheon). More industry-oriented than MIT or Caltech but not at the expense of research depth.

NC State University — Department of Mechanical and Aerospace Engineering

Best for: Hypersonics, computational aerodynamics, high-speed propulsion

NC State's AE program is aeronautics-focused, not astronautics. Its genuine national strength is hypersonics — well-funded, growing, and anchored by real facility investment. The space systems side is thin. If hypersonics or high-speed computational aerodynamics is the direction, NC State deserves serious consideration. If electric propulsion or astrodynamics is the priority, look elsewhere.

Key faculty:

- Venkat Narayanaswamy — Hypersonic aerothermodynamics, scramjet inlet design, plasma diagnostics in hypersonic shock layers. Built the first Mach 6 long-duration hypersonic wind tunnel in the Southeast (\$2.4M in DoD facility grants). One of eight national teams selected by the Joint Hypersonics Transition Office. The most externally funded aerospace researcher in the department.
- Jack Edwards — Computational hypersonics, CFD for scramjet and ramjet propulsion, reacting flows at hypersonic speeds. NASA and DoD funded. Endowed professorship (Angel Family Distinguished Professor).
- Laura Paquin — High-enthalpy boundary layer transition, femtosecond laser diagnostics, hypersonic aerothermodynamics. Former Karles Fellow at the Naval Research Laboratory. Building her HEAT Lab at NC State — still early stage but well-credentialed.
- Hong Luo — Computational fluid dynamics, computational magnetohydrodynamics (MHD), high-performance computing. His MHD work is plasma-adjacent.
- André Mazzoleni — Space systems, orbital mechanics (coursework), conceptual planetary exploration systems including a solar wind electric sail concept. Limited active funded research in traditional astrodynamics — better as a course instructor than a research supervisor for GNC or trajectory work.

What's not here: No electric propulsion faculty, no Hall thruster lab, no plasma propulsion experimental work. No active funded astrodynamics research program. Students targeting EP should go to Michigan, Princeton, or Georgia Tech. Students targeting astrodynamics should go to CU Boulder, Purdue, or UT Austin.

PhD funding: generally offered to admitted students via RA/TA appointments. Priority deadline December 15. MS students are typically unfunded.

Notes: NC State is in the Research Triangle (Raleigh) — a genuine tech hub with RTP nearby. Affordable, good climate, strong university culture. The hypersonics program is DoD-connected and growing. For a student with a plasma physics background willing to redirect toward hypersonic aerothermodynamics rather than EP, the plasma diagnostic work in Narayanaswamy's lab is a legitimate bridge.

University of Texas at Austin — Dept of Aerospace Engineering and Engineering Mechanics

Best for: Astrodynamics, GNC, computational methods, hypersonics

UT Austin has a strong astrodynamics group and growing computational AE. The Texas Advanced Computing Center (TACC) provides significant computational resources. Austin is the home of SpaceX's Starship development site nearby.

Key faculty:

- Maruthi Akella — Astrodynamics, orbital mechanics, spacecraft GNC
- Todd Humphreys — Navigation, GPS, autonomous systems
- Philip Varghese — Rarefied gas dynamics, hypersonics

Notes: Austin has become expensive but remains more affordable than coastal cities. Strong tech industry ecosystem. UT Austin AE is large and well-funded.

International programs

TU Delft — Faculty of Aerospace Engineering (Netherlands)

Best for: Propulsion, space systems, structures, wind energy, broad AE

TU Delft's aerospace faculty is one of the best in Europe — genuinely world-class in several areas, not merely a European consolation prize. Strong in electric propulsion, spacecraft systems, and structures. As an EU citizen, Olivia would pay EU tuition rates — dramatically lower than US programs — and is eligible for Dutch and EU scholarships.

Key faculty / groups:

- Space Propulsion group — Electric propulsion, green propellants
- Astrodynamics and Space Missions group — Orbital mechanics, mission design
- Structures and Composites — Lightweight aerospace structures

Notes: Delft is a small, beautiful Dutch city near The Hague and Rotterdam. The faculty is international and English-language research is standard. Very high quality of life; Dutch salaries and living costs are moderate by Western European standards. Strong industry connections to Airbus, ESA, NLR.

ISAE-SUPAERO — Institut Supérieur de l'Aéronautique et de l'Espace (France)

Best for: Broad AE, space systems, propulsion, French/European space industry pipeline

One of the most prestigious aerospace engineering schools in the world, period. Strong ties to CNES (French space agency), Airbus, Thales. Located in Toulouse — France's aerospace capital.

Notes: Programs increasingly available in English. Tuition is very low for EU students. Toulouse is a genuinely pleasant city. The SUPAERO degree carries significant weight in European and international aerospace. Worth considering if European career trajectory is acceptable.

KTH Royal Institute of Technology — Stockholm (Sweden)

Best for: Space systems, propulsion, computational methods, Scandinavian career access

KTH is Sweden's leading technical university, strong in space systems and related engineering. As a Danish EU citizen, essentially free tuition. Stockholm is one of the world's most livable cities. The Scandinavian aerospace industry (Saab, RUAG, OHB) is accessible from here.

Notes: Research intensity is somewhat lower than top US programs but the environment is excellent and the cost of entry near zero. A KTH master's followed by a US PhD is a credible path.

Imperial College London — Department of Aeronautics

Best for: Structures, computational aerodynamics, propulsion, broad AE

Imperial is one of the UK's top engineering schools with a strong aeronautics department. Post-Brexit, Danish students are international (non-EU), which means higher fees than pre-2021 — less favorable than TU Delft or KTH.

Notes: London is expensive. Imperial's research is strong; the location provides access to the UK aerospace industry (Rolls-Royce, BAE Systems, Airbus UK). Worth considering if the UK track appeals.

Propulsion focus — if rockets are the draw

Propulsion splits into two distinct worlds with different research cultures, different physics, and different relevance to a physics background. Knowing which one pulls matters before building a list.

Chemical propulsion — combustion, thrust, launch vehicles

The traditional rocket engine. Fuel and oxidizer combust to produce thrust. This is what lifts things off the ground. The physics is thermodynamics, combustion chemistry, fluid dynamics at high temperatures and pressures. A physics background helps with the fundamentals but the domain knowledge — combustion instability, injector design, nozzle flow — is engineering-specific.

Purdue — Zucrow Laboratories

The definitive address for chemical rocket propulsion in US academia. Named after Maurice Zucrow — the "father of American rocketry" — these labs have a direct lineage from the early US space program. High-pressure combustion cells, liquid rocket test stands, solid rocket facilities. More rocket test infrastructure than almost anywhere outside NASA and the major primes. Faculty: Stephen Heister (liquid propulsion, combustion instability), Alina Alexeenko (rarefied gas dynamics, microthrusters). If Raptor-level combustion engineering is the interest, Purdue is the address.

Georgia Tech — Ben T. Zinn Combustion Lab

Vigor Yang is one of the most prominent researchers in liquid rocket combustion in the country. Also has rotating detonation engines (James Braun, BEFAST Lab) — a potential replacement for conventional combustion cycles, active DoD interest. Strong industry connections to launch vehicle companies.

Penn State — Propulsion Engineering Research Center

Solid and liquid propulsion. Less prominent nationally than Purdue or Georgia Tech but a real, funded program.

University of Alabama Huntsville (UAH)

Sits next to NASA Marshall Space Flight Center — where Saturn V engines were developed and where SLS propulsion now lives. UAH has research ties to Marshall that no other US university can match by proximity. A smaller institution but a unique gateway into NASA's propulsion ecosystem. Worth knowing if NASA career trajectory is the goal.

Electric propulsion — plasma, ions, Hall thrusters

Spacecraft thrusters that use electric fields to accelerate ions. No combustion. Far more efficient than chemical propulsion in deep space. The physics is plasma physics, electromagnetism, and gas dynamics at low pressure — directly continuous with a theoretical physics background. This is the area where coming from plasma physics is an asset rather than a gap.

Michigan — Plasmadynamics and Electric Propulsion Lab (PEPL)

One of the two or three best EP labs in the country. Hall thrusters, ion engines, Faraday probes, plasma plume diagnostics. Benjamin Jorns leads the lab now — his work on oscillation dynamics in Hall thrusters is among the most cited in the field. Ann Arbor is affordable; the lab is well-funded; the program is strong.

Princeton — Electric Propulsion and Plasma Dynamics Lab (EPPDyL)

Founded by Edgar Choueiri, one of the field's defining figures. Choueiri recently retired but the lab continues. World-class reputation. The physics culture at Princeton bleeds across from the physics department — a genuinely good environment for someone coming from that background.

Georgia Tech — High Power Electric Propulsion Lab (HPEPL)

Mitchell Walker runs the largest EP test facility in US academia. Focus on high-power Hall thrusters for deep space and nuclear-electric propulsion — the next generation of propulsion for missions to Mars and beyond. Georgia Tech appears on both the chemical and electric propulsion lists; its propulsion program is the broadest in the country.

Colorado State — Center for Electric Propulsion and Power Engineering (CEPPE)

Solid EP research group, less prominent than Michigan or Princeton but genuine. Fort Collins is affordable and livable.

MIT — Space Propulsion Lab

Paulo Lozano works on miniaturized electric propulsion — electrospray thrusters for CubeSats, ion engines at small scale. Cutting edge but small-scale; if large-system EP is the interest, Michigan or Georgia Tech are better fits.

Which propulsion world fits this profile

	Chemical	Electric
Core physics	Thermodynamics, combustion	Plasma physics, electromagnetism
Olivia's background overlap	Indirect	Direct
Engineering gap to bridge	Thermodynamics, chemistry	Smaller — plasma physics transfers
Industry connections	SpaceX, Blue Origin, Aerojet	Satellite industry, deep space programs
Best programs	Purdue, Georgia Tech, UAH	Michigan, Princeton, Georgia Tech

Georgia Tech is the only program that appears prominently on both lists. If the choice between chemical and electric isn't made yet, Georgia Tech gives the most room to explore and decide. The Ben T. Zinn Combustion Lab and the HPEPL are both there.

Progr	Location	Best for	Profile fit
MIT Aero Astro	Boston MA	Electric propulsion, space systems, astrodynamics	Very strong

Princeton	Princeton NJ	Electric propulsion (world-class), plasma	Very strong
Michigan	Ann Arbor MI	Electric propulsion, space weather	Very strong
CU Boulder	Boulder CO	Astrodynamics (exceptional), space environment	Very strong
Caltech	Pasadena CA	Fundamental propulsion, fluids	Strong, selective
Purdue AAE	W. Lafayette IN	Broad propulsion, astrodynamics	Strong
Georgia Tech	Atlanta GA	Hypersonics, broad AE, industry	Strong
NC State MAE	Raleigh NC	Hypersonics, computational aero	Strong for hypersonics; no EP
UT Austin	Austin TX	Astrodynamics, GNC, computation	Strong
TU Delft	Delft NL	Broad AE, EU tuition, world-class	Strong, affordable
ISAE-SUPA	Toulouse FR	Broad AE, European pipeline	Strong, European
KTH	Stockholm SE	Space systems, Scandinavian access	Good, near-free
Imperial	London UK	Structures, UK industry	Good, expensive

See also: [landscape.md](#) for sub-field descriptions, [application-guide.md](#) for how to apply